

SQL Data Warehouse Project

Bronze → Silver → Gold | Medallion Architecture

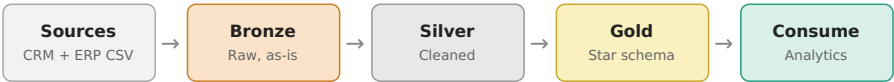
SQL Server • ETL • Star Schema • Analytics

An end-to-end data warehouse consolidating CRM and ERP data into a clean, business-ready star schema — built to practice real-world data engineering: ETL pipelines, data cleaning, dimensional modeling, and SQL analytics.

Prepared by Dania

Architecture

Raw CRM and ERP data flows through three progressively cleaner layers before reaching reporting.



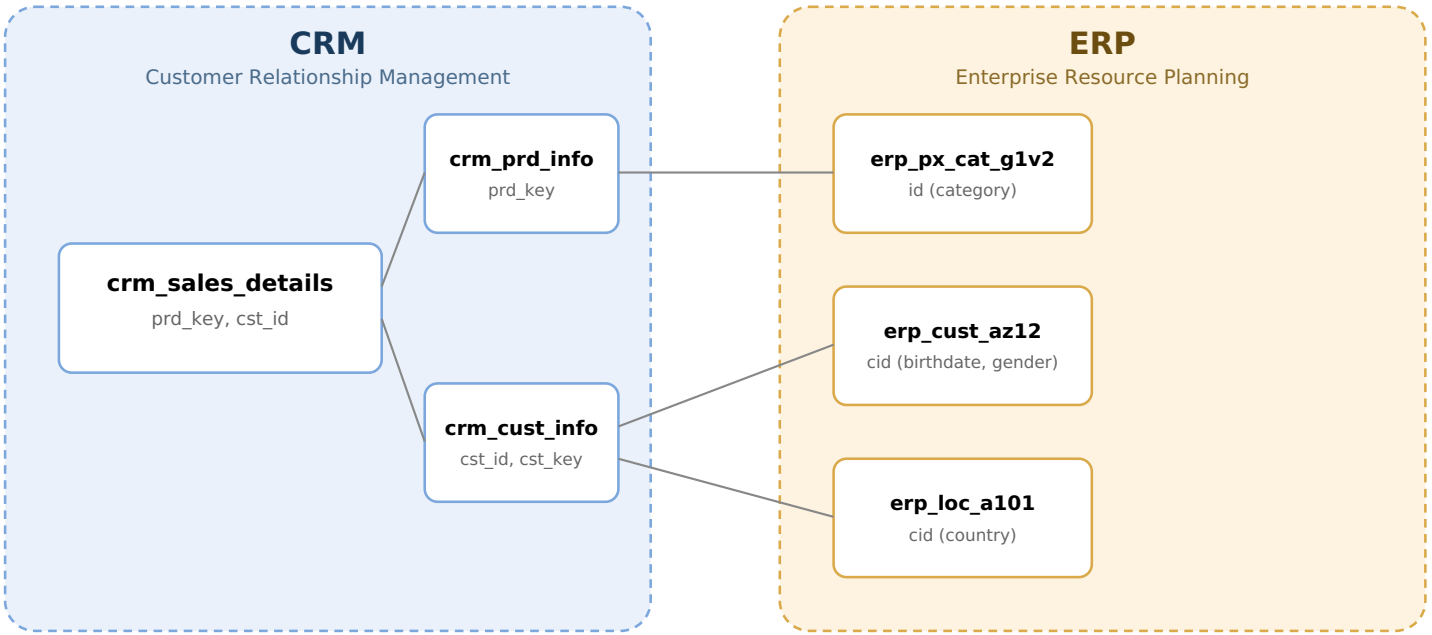
	Bronze	Silver	Gold
Contains	Raw, unprocessed data	Clean & standardized data	Business-ready views
Object type	Tables	Tables	Views
Transformations	None	Cleansing, dedup, standardization	Integration, star schema

What I built

- Stored procedures (load_bronze, load_silver) with logging, automating the full pipeline
- Data cleaning: trimming, code standardization, deduplication with ROW_NUMBER ()
- Recalculated invalid sales values and derived product end-dates with LEAD ()
- Validated the model with uniqueness and referential integrity checks

Data integration — how tables relate

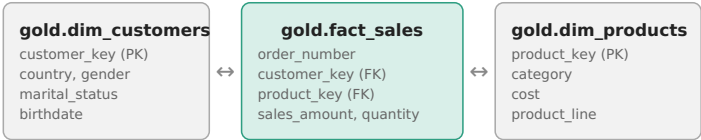
CRM and ERP are separate systems tracking overlapping information about the same customers and products. Integration means joining them on shared keys so one clean view results.



CRM table	Joined with	Adds
crm_cust_info	erp_cust_az12	Birthdate, gender (fallback)
crm_cust_info	erp_loc_a101	Country / location
crm_prd_info	erp_px_cat_g1v2	Category, subcategory

Data model (star schema)

Gold layer: one fact table connected to two dimension tables via surrogate keys — built for fast, flexible analytical queries.



Sales rule: sales_amount = quantity × price, recalculated in Silver whenever the source value was missing or inconsistent.

Data catalog — Gold layer

Table	Role	Key columns
gold.dim_customers	Dimension	customer_key, first_name, last_name, country, gender, marital_status, birthdate
gold.dim_products	Dimension	product_key, product_name, category, subcategory, cost, product_line
gold.fact_sales	Fact	order_number, customer_key (FK), product_key (FK), order_date, sales_amount, quantity, price

SQL concepts practiced

Stored procedures BULK INSERT TRY/CATCH CTEs Window functions RANK() Surrogate keys Star schema design Data quality checks

Built while following the SQL Data Warehouse Project tutorial by Data With Baraa, then extended with original analytics queries and this documentation.